

AI LAB: COMPUTER VISION AND NLP

From Fundamentals to Modern Deep Learning Techniques

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PROMETHEUS LABORATORY

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TELEGRAM GROUP

https://t.me/+MEVU_BF-V6UyY2M0

CLASSROOM

Code: vyoirmup

Link: <https://classroom.google.com/c/ODM1OTc0MzU2MjYx?cjc=vyoirmup>

Academic Year 2025/2026



SAPIENZA
UNIVERSITÀ DI ROMA

INFORMATION ON THE COURSE ORGANIZATION



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Teaching Staff and Main Topics Covered

Topics Covered by Each Teacher



Danilo Avola (Department of Computer Science, Sapienza University of Rome - Co-founder of Prometheus Lab (Italy))

- **Computer Vision Pipeline:** acquisition, preprocessing, features, training, evaluation, deployment;
- **Core tasks:** image classification and semantic segmentation with practical examples, image segmentation;
- **Model Design Choices:** architectures, losses, metrics, and common training challenges and mistakes (some precautions);
- **Advanced Topics:** saliency detection and change detection for real-world scenarios in complex and dynamic visual environments;
- **State-Of-The-Art Directions:** anomaly detection and robustness in complex environments with noisy data and unexpected domain shifts.



Daniele Pannone (University of International Studies of Rome – Co-Founder of Prometheus Lab (Italy))

- **Foundations of Digital Images:** sampling, resolution, color spaces, and transformations for robust acquisition and visual representation;
- **Neural Network Basics:** perceptrons, CNN intuition, optimization and regularization with practical intuition, examples, and design trade-offs;
- **Machine Learning Recap:** supervised learning workflow, validation and generalization including modeling, bias-variance, and error analysis;
- **Practical Deep Learning with PyTorch:** tensors, datasets, training loops, debugging plus GPU usage, checkpoints, and logging best practices;
- **Hands-on Modeling Practice:** experimentation, hyperparameters, and reproducible results using clean code, seeds, and proper reporting.



Mario Lezochè (Professor at University of Lorraine/IUT Epinal – Co-Founder of Prometheus Lab (France))

- **Introduction to NLP:** what language data is and how we represent text;
- **Text Preprocessing Basics:** tokenization, normalization, and simple feature extraction;
- **Core NLP Tasks Overview:** classification, sentiment analysis, and information extraction;
- **Modern NLP Intuition:** embeddings, attention, and the idea behind Transformers.
- **Practical NLP Workflow:** datasets, training, evaluation, and common baseline models.



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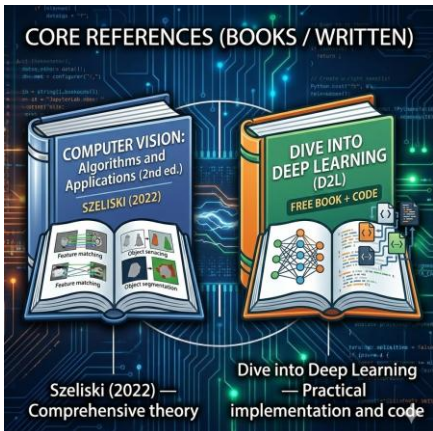
Useful Resources and Platforms for the Course

Platforms & Tools (Setup)



- **Python 3.12 (Anaconda/Miniconda):** offers broad ML library support with minimal installation issues.
- **Python Packages:** numpy, pandas, matplotlib, scikit-learn, tqdm.
- **PyTorch 2.7.0 (latest stable):** deep learning framework for training/deploying neural networks (CPU/GPU).
- **VS Code 1.109 (latest stable):** lightweight IDE with debugging, Git integration, and extensions.
- **JupyterLab 4.5.5 (latest stable):** interactive notebooks for experimentation, visualization, and debugging.
- **OpenCV (pip install opencv-contrib-python):** core CV library for image/video and processing pipelines.
- **Recommended Git/GitHub:** version control and collaboration for reproducible projects and assignments.
- **Recommended Google Colab:** cloud notebooks with GPU support for heavy training or limited hardware.
- **Optional:** FFmpeg, TensorBoard, Weights & Biases.

Books & Reading Materials (Free Resources)



- **Szeliski (2022) - Computer Vision: Algorithms and Applications (2nd Ed.)**
<https://szeliski.org/Book/download.php>
- **Dive into Deep Learning (D2L) - Free Deep Learning Book + Code**
<https://d2l.ai/>
- **PyTorch Official Tutorials - Free, Practical, Continuously Updated**
<https://pytorch.org/tutorials/>
- **Speech and Language Processing (Jurafsky & Martin) – Free Online + PDF**
https://web.stanford.edu/~jurafsky/slp3/?utm_source=chatgpt.com
- **MIT 6.S191 - Introduction to Deep Learning**
<https://introtodeeplearning.com/>



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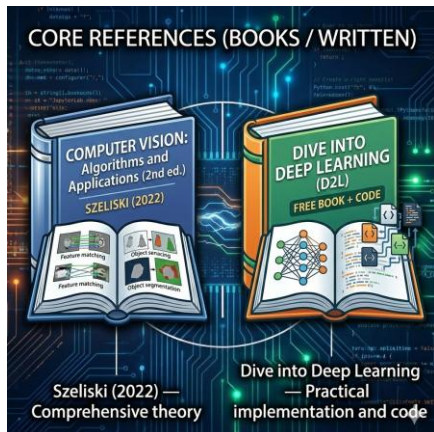
Useful Resources and Platforms for the Course

Video Resources (YouTube)



- **FreeCodeCamp - PyTorch Full Course:** A comprehensive, beginner-friendly PyTorch course covering tensors, datasets, model building, and training loops. It includes practical examples for both deep learning and classic machine learning workflows, with a strong hands-on focus and end-to-end projects:
https://www.youtube.com/watch?v=V_xro1bcAuA
- **Playlist (Extra):** A GAN-focused playlist covering Generative Adversarial Networks, from introductory concepts to practical PyTorch implementations. Useful as an optional deep dive into generative modeling.
<https://www.youtube.com/playlist?list=PLhhyoLH6ljfwlp8bZnzX8QR30TRcHO8Va>

Additional Materials (Free Resources)



- **PyTorch Course Notebooks/Repo:**
<https://github.com/mrdbourke/pytorch-deep-learning>
- **Practical NLP/Transformers: Hugging Face course**
https://huggingface.co/learn/llm-course/en/chapter1/1?utm_source=chatgpt.com
- **Deep Learning (Goodfellow, Bengio, Courville) - Free Online Textbook**
<https://www.deeplearningbook.org/>
- **Stanford CS229 — Machine Learning Lecture Notes (PDF, Free)**
https://cs229.stanford.edu/main_notes.pdf?utm_source=chatgpt.com



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Useful Resources and Platforms for the Course

Exam Format



- Students must form **groups of 1 to 4 members**, depending on the project scope, complexity and team preference.
- Each group must propose and develop a **research-oriented project**, grounded in the course topics, focused on **Computer Vision (CV)**, **Natural Language Processing (NLP)**, or more generally on **Signal Analysis** (e.g. Wi-Fi Signals, EEG Signals, ECG Signals, EMG Signals, IMU Signals, EDA/GSR Signals, etc...).
- Both **Single-Modality Projects** (e.g., Computer Vision only; Natural Language Processing only, Wi-Fi only; EEG only) and **Multimodal Projects** (e.g., Computer Vision + Audio; Audio + EEG; Natural Language Processing + EEG, EEG + ECG + EDA, or other modality combinations) are allowed.
- To complete the exam, each group must:
 - Develop a **working prototype/demonstrator (code)** that is well-commented, organized, and reproducible;
 - Write a **technical report in the style of a scientific paper**, clearly describing **the problem, state of the art, innovative/alternative contribution, methodology, experiments, and results**;
 - Prepare a **PowerPoint presentation** that effectively summarizes the report (paper) and highlights the key contributions of the proposed project with respect to the current state of the art.

During the **oral exam**, **individual questions** may be asked to each group member to assess **their personal understanding, role, and contribution to the project**.



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How To Take The Exam



- Each group member must register on **Infostud** for the exam date.
- One group member must submit, via an online form (which will be shared on Google Classroom and in the Telegram group), **at least one week before the exam date**, a **ZIP file** containing all project materials: **the Code, the Report (i.e., Scientific Paper), and the PowerPoint presentation.**
- The **ZIP file** must be named as follows: “**exam date (YYYY-MM-DD) – project acronym – participants’ surnames separated by a plus sign**”. Example:
 - **2026-06-10---SEARCHER---Avola+Pannone+Lezochè.zip**
 - Please keep spaces, hyphens, and the plus sign exactly as shown in the example.
- Inside the **ZIP file**, the following files must be included, continuing with the same example:
 - SEARCHER-(Code).zip;
 - SEARCHER-(Paper).pdf;
 - SEARCHER-(Presentation).pptx.
 - Note: The template for the Report (i.e., Scientific Paper) will be provided later on Google Classroom and in the Telegram group.
- **The final grade will depend on several factors, including:**
 - Project success and overall complexity;
 - Level of novelty, originality, innovation, and technical ambition;
 - Quality of the answers to individual oral questions;
 - Quality of the code, the scientific paper, and the presentation.